

Card sort — could we get it back?

Cut along the lines. One set per pair. Keep the sets bagged — they are used again in Lesson 4.

Two piles Could get it back — you could return it to what it was. | Gone for good — there is no way back. If you cannot decide, start a third pile in the middle. “Not sure” is a useful answer, not a failure.

Ice cubes left on the bench	Salt stirred into warm water
A banana skin going brown	Chocolate melting in your hand
A candle burning down	Wet clothes drying on the line
Bread being toasted	A nail left outside for a year
Water boiling in a kettle	Fireworks going off
Butter melting in a pan	Concrete setting

Card sort — eight cards

The same task with the four hardest cards removed. Same two piles.

Ice cubes left on the bench	Salt stirred into warm water
A banana skin going brown	Chocolate melting in your hand
Wet clothes drying on the line	Bread being toasted
A nail left outside for a year	Water boiling in a kettle

Name	Class	Date
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1	Ice melting	Is the water that is left the same stuff as the ice?
What I observed Could we get it back? YES / NO / NOT SURE The one piece of evidence that decided it:		

2	Salt dissolving	Where has the salt gone? Is it still in there?
What I observed Could we get it back? YES / NO / NOT SURE The one piece of evidence that decided it:		

3	Steel wool in vinegar	What has appeared that was not there this morning?
What I observed Could we get it back? YES / NO / NOT SURE The one piece of evidence that decided it:		

4	Bicarbonate and vinegar	<i>The bubbles are made of something. Where did it come</i>
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		from?
What I observed Could we get it back? YES / NO / NOT SURE The one piece of evidence that decided it:		

5	Candle burning (teacher)	Is this one change or two?
What I observed Could we get it back? YES / NO / NOT SURE The one piece of evidence that decided it:		

6	Bread — untoasted, toasted, burnt	At which slice did it stop being bread?
What I observed Could we get it back? YES / NO / NOT SURE The one piece of evidence that decided it:		

Push me

- Find a station that belongs in BOTH categories. Which one, and what is your argument?
- Rank the six from most certain to least certain. What made the uncertain ones uncertain?

Station circuit — tick what you see

Name	Class	Date
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Tick every box that matches what you see at the bench. Then write one reason.

Bench	Changed colour	Made bubbles or gas	Got hot or cold	Smells different	New solid appeared	Cannot be undone	Get it back?
1 • Ice melting							Y / N
2 • Salt dissolving							Y / N
3 • Steel wool in vinegar							Y / N
4 • Bicarbonate and vinegar							Y / N
5 • Candle burning (teacher)							Y / N
6 • Bread — untoasted, toasted, burnt							Y / N

Now pick just one bench

Which bench was it? _____

Write one reason for the decision you made there:

Setting up the rust investigation

One sheet per group. These jars sit on the shelf for five weeks — you do not get to start again.

Group	
Our condition	
Date set up	

Our prediction

How much rust will your condition make compared with everyone else's — most in the room, least, or somewhere in the middle? Say why.

Steps

1. **Check your jars.** Any chips or cracks, swap it.
2. **Mark each nail** halfway along with the marker. That line is what “half covered” means — not the jar.
3. **Weigh 3 nails together.** Write the number in the table below. Do this for each jar.
4. **Put 3 nails in each jar.**
5. **Set up your condition** in all 3 jars, the same way every time. Half covered = water to the marker line. Fully covered = at least 1 cm over the nail.
6. **Draw a line on the outside** of each jar where the water comes to. You top it back up to that line every week.
7. **Fill in and cut off the labels** at the bottom. Stick one on each jar.
8. **Lid on or off?** Only Tuesday A and F get a lid. Everything else stays open.
9. **Write the repeat sentence.** Then put the jars on the shelf.

Why three jars and not one If you only had one and something went wrong with it, you would never know. Three lets you spot the odd one out.

Starting mass

	Jar 1	Jar 2	Jar 3
Mass of 3 nails (g)			

Record to the precision your scales actually give you. If the scales read whole grams, do not write decimal places — that is an error, not extra care.

The repeat sentence

Write the ONE sentence that would let another group set this up exactly the same way, without asking you anything. Name the nails, the liquid, how much, and whether the lid is on.

Check it against this: “Three 65 mm nails, half covered in tap water, lid off, on the shelf by the window.” Does yours give that much?

Cut off these labels and stick one on each jar

RUST INVESTIGATION Group: _____ Condition: _____ Jar size: _____ Jar 1 of 3 Nail mass: _____ g Date: _____	RUST INVESTIGATION Group: _____ Condition: _____ Jar size: _____ Jar 2 of 3 Nail mass: _____ g Date: _____	RUST INVESTIGATION Group: _____ Condition: _____ Jar size: _____ Jar 3 of 3 Nail mass: _____ g Date: _____
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Jar G group You have a fourth jar. Write your own label for it: Jar 4 of 4, condition G, and the mass of the three ordinary nails only.

Safety Glasses on. Nails counted out and counted back in. Jars stay on the bench and are carried with two hands, one at a time. Report any spill straight away and clean it up.

Exit tickets — cut into six

Name: _____ Which station was hardest to classify? _____ What would settle it? _____ _____	Name: _____ Which station was hardest to classify? _____ What would settle it? _____ _____
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Rust coverage scale and weekly log

Laminate the scale and keep it with the jars. Five minutes at the start of each lesson: photograph, rate, record. No discussion.

Rating the coverage

Score	Name	What you can see
0	None	Nail looks the same as the day it went in. Still shiny.
1	Traces	A few orange specks or a faint tinge. You have to look for it.
2	Patchy	Obvious orange patches, but plenty of clean metal still showing.
3	Mostly covered	More rust than clean metal. Only small shiny areas left.
4	Fully covered	No clean metal visible. Rust may be flaking off into the liquid.

Rate all three jars separately Do not average them in your head. If your three jars disagree, that is data — write all three down and think about why they differ.

Weekly log

Week	Date	Jar 1	Jar 2	Jar 3	Anything you notice
Week 1 (set up)					
Week 2					
Week 3					
Week 4					
Week 5 (final)					

Week 5 only — the weigh-in

Take the nails out with tongs. Rinse, blot, and dry them properly before weighing — water left on a nail reads as rust and will invent a result that is not there.

	Jar 1	Jar 2	Jar 3
Starting mass (g)			
Final mass (g)			
Difference (g)			

If the nails are heavier than when they went in — where did the extra mass come from?

